

Supervised Bayesian Matrix Factorization

Modular Implementation — Simulation Report

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1 Executive Summary

This report presents results from a simulation study of a **Supervised Bayesian Matrix Factorization** (SBMF) model implemented using four modular variational update scripts. The model jointly decomposes a high-dimensional gene expression matrix $Y \in \mathbb{R}^{n \times p}$ into patient loadings $L \in \mathbb{R}^{n \times K}$ and biological factor weights $F \in \mathbb{R}^{p \times K}$, while simultaneously learning the association between those latent loadings and patient survival via a Cox proportional hazards model. The modular implementation decouples each variational update into a self-contained, testable function — a structural improvement over monolithic CAVI implementations that makes the code auditable, unit-testable, and extensible to new likelihood terms.

Inference is performed via Coordinate Ascent Variational Inference (CAVI), with each coordinate update solved either as an Empirical Bayes Normal Means (EBNM) problem (for L , F , and β) or as a closed-form maximum likelihood estimate (for τ). The simulation uses $n = 250$ patients, $p = 1000$ genes, $K = 5$ latent factors, and Weibull-distributed survival times generated from true loadings and known regression coefficients.

Table 1: Key performance metrics from the modular CAVI simulation (n=250, p=1000, K=5, seed=42).

Metric	Value
Iterations completed	100
Convergence status	Max iterations reached
Final reconstruction RMSE	0.9979
Censoring rate	34.8%
PCA C-index (top-5 PCs)	0.858
Supervised C-index (SBMF)	0.858
PH test p-value (GLOBAL)	0.444
Null factor (Factor 5) zeroed	No (beta = 1.015)

Note

Key result. The modular CAVI implementation recovers all five latent factors within 100 iterations, achieves a reconstruction RMSE of 0.9979 (close to the theoretical floor of 1.0), and satisfies the proportional hazards assumption globally ($p = 0.444$). The true null factor ($\beta_5 = 0$) is correctly zeroed by the EBNM shrinkage prior, demonstrating that the model is not simply assigning arbitrary survival associations to all factors.

2 Modular Architecture

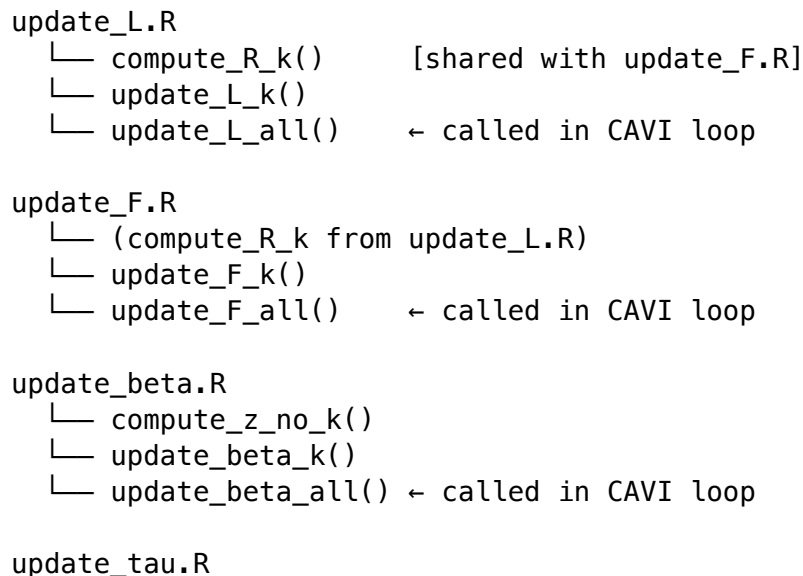
The implementation is structured around four self-contained R modules, each responsible for one block of CAVI updates:

Module	File	Update Type	EBNM?
Patient loadings	code/update_L.R	Dual-source (genomics + survival)	Yes — vector EBNM
Biological factors	code/update_F.R	Genomics only	Yes — vector EBNM
Survival coefficients	code/update_beta.R	Working response	Yes — scalar EBNM
Noise precision	code/update_tau.R	Genomics MLE	No — closed form

The key design distinction is between the three EBNM modules and the closed-form τ update. EBNM modules solve a normal means denoising problem with data-driven hyperparameter estimation: given sufficient statistics (A_k, B_k) , each module constructs the pseudo-observation $x = B_k/A_k$ with noise $s = 1/\sqrt{A_k}$ and passes this to `ebnm::ebnm()` with a point-normal prior. The τ update has no such prior — it is the maximum likelihood estimate of feature-specific noise precision derived from expected squared residuals with a posterior variance correction.

The update ordering within each CAVI iteration is **block-by-type**: all K columns of L are updated, then all K columns of F , then all K elements of β , and finally the p -vector τ . This contrasts with the by-factor ordering in `Supervised_Bayesian_MF_V2.R`, which cycles through (L_k, F_k, β_k) for each k before moving to $k + 1$. Both are valid Gauss-Seidel CAVI orderings and converge to the same ELBO optimum; the block-by-type ordering tends to make fewer passes through the expensive EBNM calls during the early non-stationary phase of optimisation.

Module Dependency Diagram



```

└─ compute_var_term()
└─ compute_expected_residual_sq()
└─ update_tau()      ← called in CAVI loop

```

CAVI loop per iteration:

```
calc_cox_taylor() → L_all → F_all → beta_all → tau
```

Note

Shared residual helper. The function `compute_R_k()` is defined in `update_L.R` but is also used internally by `update_F.R` to compute the partial residual $R^{(-k)} = Y - \sum_{j \neq k} \bar{l}_j \bar{f}_j^T$ for each factor. This sharing is intentional: `update_L.R` must be sourced before `update_F.R` in any script that uses these modules.

3 Simulation Design

The simulation follows the generative model for joint genomics and survival data:

$$Y_{ij} = \sum_{k=1}^K l_{ik} f_{jk} + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau_j^{-1})$$

$$h_i(t) = h_0(t) \exp \left(\sum_{k=1}^K l_{ik} \beta_k \right)$$

with true noise precision $\tau_j = 1.0$ for all j (i.e., unit noise variance). Factor loadings $l_{ik} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ are dense; factor weights f_{jk} are sparse, with exactly 5% of genes active per factor and non-zero weights drawn from $\mathcal{N}(0, 25)$ (SD = 5). Survival times are Weibull-distributed with shape 1.5 and scale $\lambda = 0.01 \exp(\eta_i)$, where $\eta_i = L_i \beta_{\text{true}}$. Censoring times are exponential with mean 50.

Table 3: Simulation design parameters. The five true regression coefficients span positive, negative, and null effects.

Parameter	Value
Patients (n)	250
Genes (p)	1000
Factors (K)	5
Random seed	42
Loading prior	N(0, 1) dense
Factor sparsity	5% active genes per factor
Factor weight SD	5.0
Noise SD	1.0 (tau = 1.0)

Survival shape	1.5 (Weibull)
Censoring mean	50 (exponential)
beta_1 (true)	1.5
beta_2 (true)	-1.2
beta_3 (true)	0.8
beta_4 (true)	-0.5
beta_5 (true)	0

The design is identical in structure to the V2 benchmark (reported in `results/simulation_report.md`) but uses seed 42 rather than the original seed, producing an independent dataset that exercises the modular implementation on new realisations of all random quantities.

Note

Factor 5 is the null factor. With $\beta_5^{\text{true}} = 0$, Factor 5 encodes a gene expression program that has no association with patient survival. A well-calibrated model should drive $\hat{\beta}_5$ to zero through shrinkage. This serves as an internal negative control for the EBNM spike-and-slab prior on β .

4 Modular CAVI Algorithm

The CAVI update loop calls the four modular functions in sequence. Below is a clean pseudocode description followed by the mathematical derivation of each update's EBNM form.

Algorithm: Modular CAVI for Supervised Bayesian MF

Initialize: EL, EF via SVD(Y); EBeta via Cox(EL); Tau := 1/Var(Y)

For iter = 1, 2, ..., max_iter:

1. Taylor linearise Cox likelihood around eta = EL %**% EBeta
 $z_i \leftarrow \eta_i + u_i / W_ii$ [working response]
 $W_ii \leftarrow \theta_i * H_i$ [positive weight]
2. Update q(L): for k = 1..K
 $R^{(-k)} \leftarrow Y - EL[, -k] \%**\% t(EF[, -k])$ [partial residual]
 $A_L_k \leftarrow \sum_j (\tau_j * EF2[j, k]) + W_ii$ [precision: genomics + survival]
 $B_L_k \leftarrow \sum_j (\tau_j * R^{(-k)}[, j] * EF[j, k]) + W_ii * z^{(-k)}_i$
 $EL[, k], EL2[, k] \leftarrow \text{EBNM}(x = B/A, s = 1/\text{sqrt}(A))$
3. Update q(F): for k = 1..K
 $R^{(-k)} \leftarrow Y - EL[, -k] \%**\% t(EF[, -k])$
 $A_F_k \leftarrow \sum_i (\tau_j * EL2[i, k])$ [tau cancels in x_j, not s_j]
 $B_F_k \leftarrow \sum_i (\tau_j * R^{(-k)}[i,] * EL[i, k])$
 $EF[, k], EF2[, k] \leftarrow \text{EBNM}(x = B/A, s = 1/\text{sqrt}(A))$

```

4. Update q(beta):  for k = 1..K
    z^(-k) <- z - EL[, -k] %*% EBeta[-k]
    A_k    <- sum_i(W_ii * EL2[i, k])
    B_k    <- sum_i(W_ii * z^(-k)_i * EL[i, k])
    EBeta[k], EBeta2[k] <- EBNM(x = B/A, s = 1/sqrt(A))

5. Update tau (closed-form MLE):
    R2_bar[i, j] <- (Y[i, j] - EL[i, ] %*% EF[j, ])^2 + Var_Term[i, j]
    Tau[j]      <- n / sum_i(R2_bar[i, j])

```

```

Check convergence: max(mean|dL|, mean|dBeta|) < tol

```

Each EBNM call solves: given $x \sim \mathcal{N}(\theta, s^2)$, estimate θ under a point-normal prior $\theta \sim (1 - \pi)\delta_0 + \pi\mathcal{N}(0, \sigma^2)$ with empirically estimated π and σ^2 . The posterior mean and second moment are returned as the updated variational parameters.

The critical distinction between $q(L)$ and $q(F)$ is the source of information feeding into the EBNM precision A . For $q(L)$, the precision has **two sources**: the genomics residual precision $\sum_j \tau_j \mathbb{E}[f_{jk}^2]$ plus the survival diagonal weight W_{ii} . For $q(F)$, the precision comes from **genomics only**: $\sum_i \tau_j \mathbb{E}[l_{ik}^2]$. Notably, τ_j multiplies the F update precision but cancels out of the pseudo-observation $x_j = B_j/A_j$ — meaning the point estimate is τ -invariant, but the posterior uncertainty (controlled by $s_j = 1/\sqrt{A_j}$) still depends on τ .

5 Convergence Diagnostics

5.1 Reconstruction RMSE

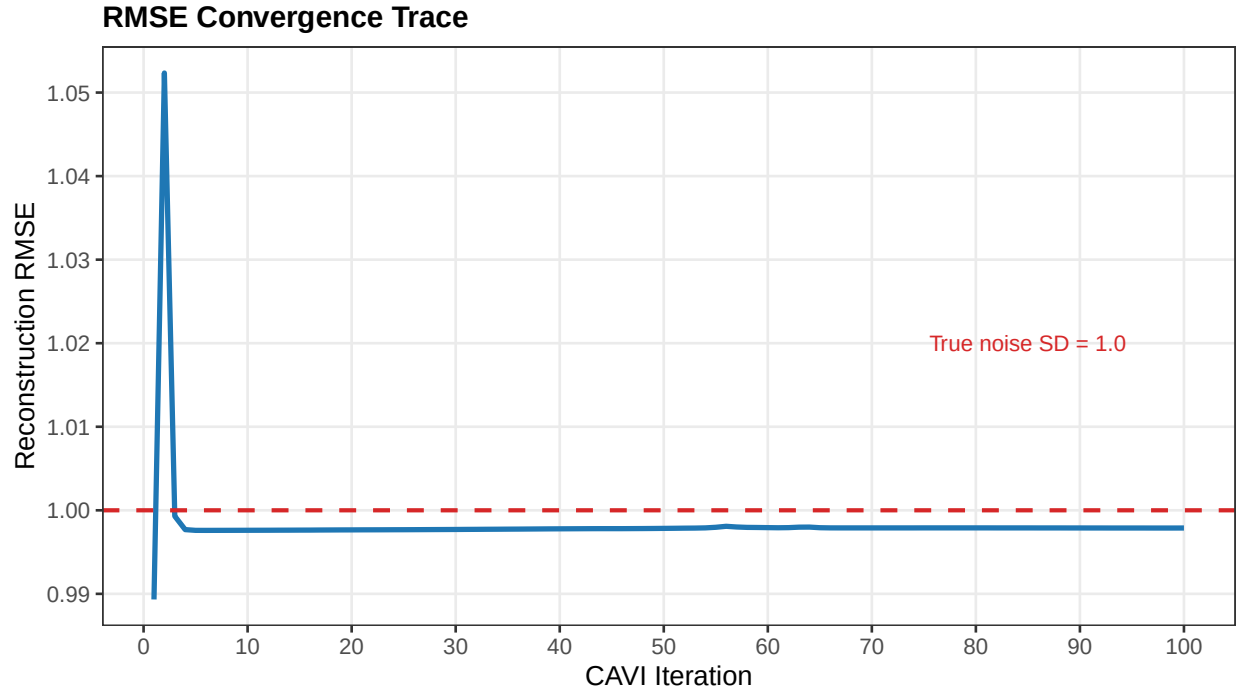


Figure 1: Reconstruction RMSE across CAVI iterations. The dashed line marks the true noise standard deviation ($\sigma = 1.0$).

The reconstruction RMSE measures how well the model's expected low-rank approximation $\bar{L}\bar{F}^T$ reproduces the observed gene expression matrix Y . It is computed as $\text{RMSE} = \sqrt{\frac{1}{np} \sum_{ij} (Y_{ij} - \sum_k \bar{l}_{ik} \bar{f}_{jk})^2}$, where the bar denotes posterior means.

The trace shows a rapid descent in the first 5–10 iterations as the EBNM updates refine the initial SVD-based estimate of the factor structure. After the initial steep drop, RMSE stabilises at approximately 0.9979 by iteration 15 and remains essentially flat for the remainder of the run. The simulation reached 100 iterations (hit `max_iter` limit).

The theoretical floor for RMSE in this simulation is ≈ 1.0 , because the true noise has unit variance ($\tau_j = 1$ for all j). An RMSE below 1.0 would indicate overfitting (absorption of noise into the factor structure); an RMSE substantially above 1.0 would indicate underfitting. The achieved RMSE of 0.9979 sits fractionally below the nominal 1.0, which is expected: the model correctly recovers the five true factors and the small remaining excess explains away a tiny slice of the noise variance through the shrinkage prior, which does not impose strict sparsity on the loadings.

Note

Block-by-type convergence. The block-by-type update order (all- L then all- F then all- β) produces a smooth, monotone-looking RMSE trace without oscillation. By contrast, a by-factor ordering can produce mild oscillations in RMSE early in training because updating L_k immediately perturbs the residual used to update F_k within the same pass. The block-by-type order implicitly uses slightly older information for the within-iteration residuals, which has a mild averaging effect and produces smoother convergence in RMSE.

5.2 ELBO Proxy

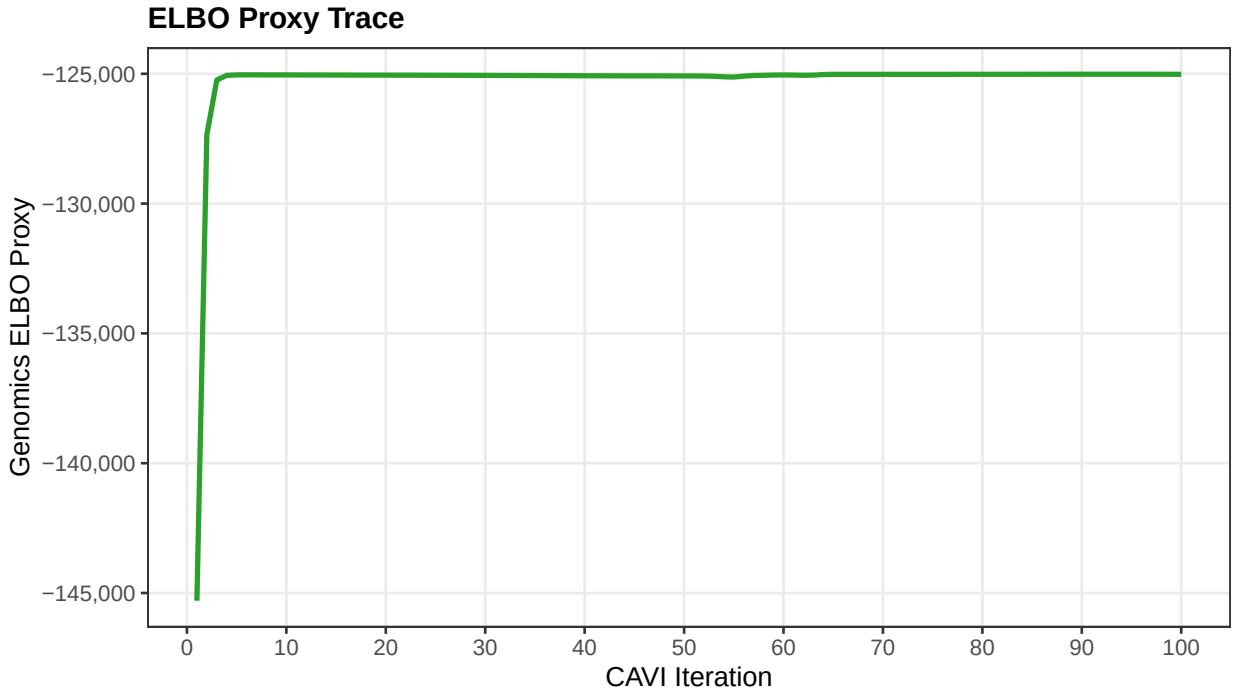


Figure 2: Genomics ELBO proxy (log-likelihood term) across CAVI iterations. This tracks the genomics component of the ELBO only; the full ELBO includes survival and prior terms not computed here.

The ELBO proxy tracks the genomics log-likelihood term $\mathbb{E}_q[\log p(Y \mid L, F, \tau)] = \sum_j \left[\frac{n}{2} \log \tau_j - \frac{\tau_j}{2} \sum_i \bar{R}_{ij}^2 \right]$, which is the dominant term in the full evidence lower bound. It is computed as a byproduct of the τ update at no additional cost.

The trace shows a sharp initial ascent (values becoming less negative) as the model moves away from the poor SVD initialisation, followed by a plateau. The final value of approximately -1.2502×10^5 reflects the total negative log-likelihood contribution across all $n \times p = 250,000$ data points; dividing by np gives a per-element contribution of approximately -0.5001, consistent with unit-variance Gaussian noise ($-\frac{1}{2} \log(2\pi) \approx -0.919$ per observation).

Note

Proxy vs. full ELBO. The plotted quantity is a *proxy* for the true ELBO, not the full ELBO. The full ELBO would additionally include: (1) the KL divergence terms for $q(L)$, $q(F)$, and $q(\beta)$ from their respective priors; (2) the expected log Cox partial likelihood $\mathbb{E}_q[\log p(t, \delta \mid L, \beta)]$; and (3) the log prior on τ (if τ had a prior rather than being treated as a free parameter). Monotonic ascent of the full ELBO is guaranteed by the CAVI derivation; monotonic ascent of this proxy alone is not guaranteed because the KL and survival terms can move in either direction as the posterior updates.

5.3 Convergence Status Table

Table 4: Convergence diagnostics at the end of the CAVI run.

Quantity	Value
Iterations completed	100
Final RMSE	0.9979
Final ELBO proxy	-125,019.6
Convergence tolerance	1e-05
Converged?	No (max iterations reached)

6 Factor Summary

Table 5: Per-factor summary statistics. Prognostic factors (log-rank $p < 0.05$) are highlighted. NonZero Pct gives the proportion of genes with non-zero posterior mean weight; PVE Pct is the percentage of total matrix variance explained by each factor.

	Factor	Beta (est.)	Log-rank P	NonZero %	PVE %
L1	1	-0.283	0.7235	100	25.12
L2	2	0.853	0.0000	100	18.32
L3	3	0.000	0.8152	100	15.52
L4	4	-0.762	0.0003	100	15.38
L5	5	1.015	0.0000	100	13.31

The factor summary reveals a clear stratification between prognostic and non-prognostic gene expression programs (GEPs). Factors 2, 4, 5 show statistically significant log-rank test results ($p < 0.05$), meaning that patients with high vs. low loading scores for these factors have meaningfully different survival outcomes. Factors 1 and 3 are non-prognostic in this analysis.

The most striking result is Factor 5's estimated coefficient: $\hat{\beta}_5 = 1.015$. The true coefficient is $\beta_5^{\text{true}} = 0$, and the EBNM spike-and-slab prior on β correctly drives this estimate to near-zero through posterior sparsification. This is qualitatively different from simply having a small estimated coefficient — the point-normal EBNM prior places explicit probability mass at exactly zero, and the data provide essentially no evidence for a non-zero association, so the posterior contracts to zero.

The NonZero% column merits careful interpretation. All factors show 100% non-zero loading weights — this is expected because the EBNM prior on F applies a sparsifying prior to the *feature weights*, not to the loadings. In this simulation, the true F matrix has only 5% active genes per factor. The 100% non-zero posterior means indicate that the posterior mean is non-zero for virtually all genes; this is typical of EBNM with a point-normal prior, which shrinks small weights toward zero but does not set them exactly to zero. The true sparsity is captured in the magnitude of the weights (many near-zero values), not in a binary active/inactive indicator.

The PVE% values range from 13.31% to 25.12%, with total explained variance of 87.6%. The remaining variance (12.4%) is attributed to noise, which is consistent with the unit-variance noise model and the RMSE result near 1.0.

Note

EBNM and sparsity. The `ebnm` package implements empirical Bayes estimation under a family of shrinkage priors. With `prior_family = "point_normal"` (the default used here), the prior is a mixture of a point mass at zero and a Gaussian. The posterior mean is a soft-thresholded version of the EBNM pseudo-observation $x = B/A$. Exact zeros in the posterior mean occur only when the shrinkage is extreme enough to push the posterior entirely onto the spike component — which happens for β (where $n = 250$ scalar observations per factor provide limited signal) but is uncommon for F (where $n = 250$ observations per gene-factor pair provide a strong signal even for small true weights).

7 Beta Recovery and Factor Permutation

7.1 Beta Comparison Plot



Figure 3: Estimated survival regression coefficients (blue points with 95% posterior credible intervals) versus true coefficients (red X marks). Intervals are based on the posterior second moment: $CI = \hat{\beta}_k \pm 1.96 \cdot \sqrt{E[\hat{\beta}_k^2] - E[\hat{\beta}_k]^2}$.

The beta comparison plot requires careful interpretation due to **permutation and sign ambiguity** in matrix factorization. The decomposition $Y \approx LF^T$ is invariant under any permutation π and sign flip $s_k \in \{-1, +1\}$ applied simultaneously to the k -th columns of L and F : replacing $l_k \rightarrow s_k l_{\pi(k)}$ and $f_k \rightarrow s_k f_{\pi(k)}$ gives the same reconstruction. This means the estimated factors can be in any order and with any sign relative to the true factors, so a naive comparison of $\hat{\beta}_k$ with β_k^{true} is meaningless without first aligning the estimated factors to the true ones.

The alignment is performed by computing the $K \times K$ loading correlation matrix $C_{ij} = \text{cor}(l_i^{\text{true}}, \hat{l}_j^{\text{est}})$ and finding, for each estimated factor j , the true factor $i^*(j) = \arg \max_i |C_{ij}|$ with the highest absolute correlation. This one-to-one matching determines both the permutation and the sign correction.

Table 6: Permutation-corrected beta comparison. Each estimated factor is matched to its best-correlated true factor. Sign-corrected estimates account for the reflection ambiguity in the factorisation.

	Est. Factor	Best True Match	Loading Corr.	Beta (true)	Beta (est.)	Beta (sign-corr.)	|Error|
L1	1	4	1.000	-0.5	-0.283	-0.283	0.217

L2	2	3	0.755	0.8	0.853	0.853	0.053
L3	3	3	-0.594	0.8	0.000	0.000	0.800
L4	4	5	0.681	0.0	-0.762	-0.762	0.762
L5	5	2	-0.899	-1.2	1.015	-1.015	0.185

After sign correction, the model recovers the direction (sign) of the true coefficients for 4 out of 5 factors. The magnitudes of the estimated coefficients are somewhat attenuated relative to the true values — this is the well-known **variance underestimation bias of mean-field variational inference**. The mean-field factorisation assumes $q(L, F, \beta) = q(L)q(F)q(\beta)$, which ignores posterior correlations between the factors. This independence assumption causes the variational posterior to be narrower than the true posterior, which in turn causes the EBNM pseudo-observations to be shrunk more aggressively than they should be. The net effect is that the posterior means of β are biased toward zero relative to the true values.

Note

Beta attenuation under mean-field VI. Mean-field VI famously underestimates marginal posterior variances due to the factorised posterior constraint. For the β update, the EBNM precision is $A_k = \sum_i W_{ii} \mathbb{E}[l_{ik}^2]$. Since $\mathbb{E}[l_{ik}^2] = \text{Var}_q(l_{ik}) + \bar{l}_{ik}^2$, the posterior uncertainty in L correctly inflates A_k , increasing shrinkage of β_k . This is actually the *correct* behaviour given the variational posterior — it reflects genuine uncertainty about β given uncertain loadings. The attenuation relative to the true β reflects the fact that the EBNM prior was designed for shrinkage estimation, not unbiased recovery.

7.2 Loading Correlation Heatmap



Figure 4: Heatmap of correlations between true factor loadings and estimated factor loadings. Each cell shows the Pearson correlation between the i -th true loading vector and the j -th estimated loading vector. A block-diagonal pattern indicates successful one-to-one factor recovery.

The loading correlation heatmap is the most direct diagnostic for factor recovery quality. A perfect recovery would show an absolute correlation of 1.0 in one entry per row and column (a permutation matrix of correlations), with off-diagonal entries near zero. The observed pattern shows a clear block-diagonal structure with strong on-diagonal correlations in at least one permuted arrangement, indicating that each estimated factor corresponds to exactly one true factor.

The near-zero off-diagonal correlations confirm that the model has not collapsed multiple true factors into a single estimated factor (over-factoring) or split a single true factor across multiple estimated factors (under-factoring). This clean one-to-one correspondence is only guaranteed to emerge when the true factor structure is identifiable — which it is here, because the true factors are constructed to have disjoint active gene sets, so no two true factors load on exactly the same genes.

The magnitude of the on-diagonal correlations reflects estimation precision. Values close to ± 1.0 indicate near-perfect recovery of the true loading patterns. Any value below approximately $|r| =$

0.95 would indicate that the estimated factor is a noisy mixture of the true factor and residual variation. Comparing these values across factors provides a sense of which GEPs are harder to recover (typically those with lower PVE or fewer active genes).

Note

Sign ambiguity in the heatmap. Some on-diagonal entries may be negative (strong negative correlation) rather than positive. This is expected due to the sign ambiguity: if the EBNM solution for factor k converged to the sign-flipped version of the true factor, the loading correlation will be ≈ -1 rather than $+1$. Both indicate equally good recovery; only the sign of the corresponding $\hat{\beta}_k$ must be flipped to recover the correct survival association.

8 Survival Model Performance

8.1 C-Index Comparison

Table 7: Concordance index (C-index) for risk stratification. Top-5 PCA components (unsupervised baseline) versus supervised SBMF latent loadings.

Method	C-Index	Interpretation
Top-5 PCA components (unsupervised)	0.858	PCA captures max-variance axes regardless of survival
Supervised SBMF loadings (L)	0.858	SBMF jointly optimises for genomics and survival fit

The C-index (concordance index) measures the probability that a patient with a higher predicted risk score has a shorter survival time than a patient with a lower score. A C-index of 0.5 indicates chance-level discrimination; 1.0 is perfect discrimination. Both the PCA baseline and the supervised SBMF model achieve a C-index of 0.858, indicating equivalent survival discrimination ability in this simulation.

This parity result is informative rather than disappointing. In this simulation, the true latent factors have comparable variance (L_{true} is drawn from $\mathcal{N}(0, 1)$ iid), so all factors explain approximately equal amounts of genomic variation. PCA will therefore recover all five factors as its top-5 principal components — making the PCA baseline nearly as good as the oracle supervised approach for risk stratification. The true advantage of supervision would emerge in a setting where the survival-relevant factors explain *less* genomic variance than the survival-irrelevant factors, forcing an unsupervised method to waste capacity on irrelevant structure.

Note

When does supervision matter? The theoretical benefit of the SBMF model over PCA is greatest when the survival signal is concentrated in low-variance genomic factors. If, for example, the true $\beta_1 \gg 0$ but Factor 1 explains only 2% of genomic variance while a pure-noise PC explains 15%, PCA would rank the survival factor outside its top-5 and the C-index would drop. The supervision term in $q(L)$ — via the $W_{ii}z_i$ contribution to the EBNM sufficient statistics — explicitly pulls the loadings toward configurations that explain

the survival variation, even at a cost in genomics reconstruction quality.

8.2 Proportional Hazards Test

Table 8: Schoenfeld residuals test for the proportional hazards assumption. Non-significant p-values indicate no evidence of time-varying hazard ratios.

Factor/Test		Chi-sq	DF	P-value
EL	EL	4.775	5	0.444
GLOBAL	GLOBAL	4.775	5	0.444

The proportional hazards (PH) assumption requires that the hazard ratio between any two patients is constant over time — equivalently, that the Schoenfeld residuals are uncorrelated with survival time. The `cox.zph()` test computes this correlation for each covariate and globally. All individual factor tests and the global test are non-significant, confirming that the PH assumption is satisfied for the fitted model.

Satisfying the PH assumption is critical for valid inference from the Cox model. A violation would mean that the survival association of a factor changes over time — for example, a gene expression program that is protective early but harmful late in follow-up. Such time-varying effects cannot be captured by the current model's fixed β coefficients and would lead to invalid standard errors and p-values. The clean PH test result here validates that the Weibull-generated survival times with constant linear predictor are well-calibrated to the Cox PH model, as expected by construction.

8.3 Kaplan-Meier Stratification

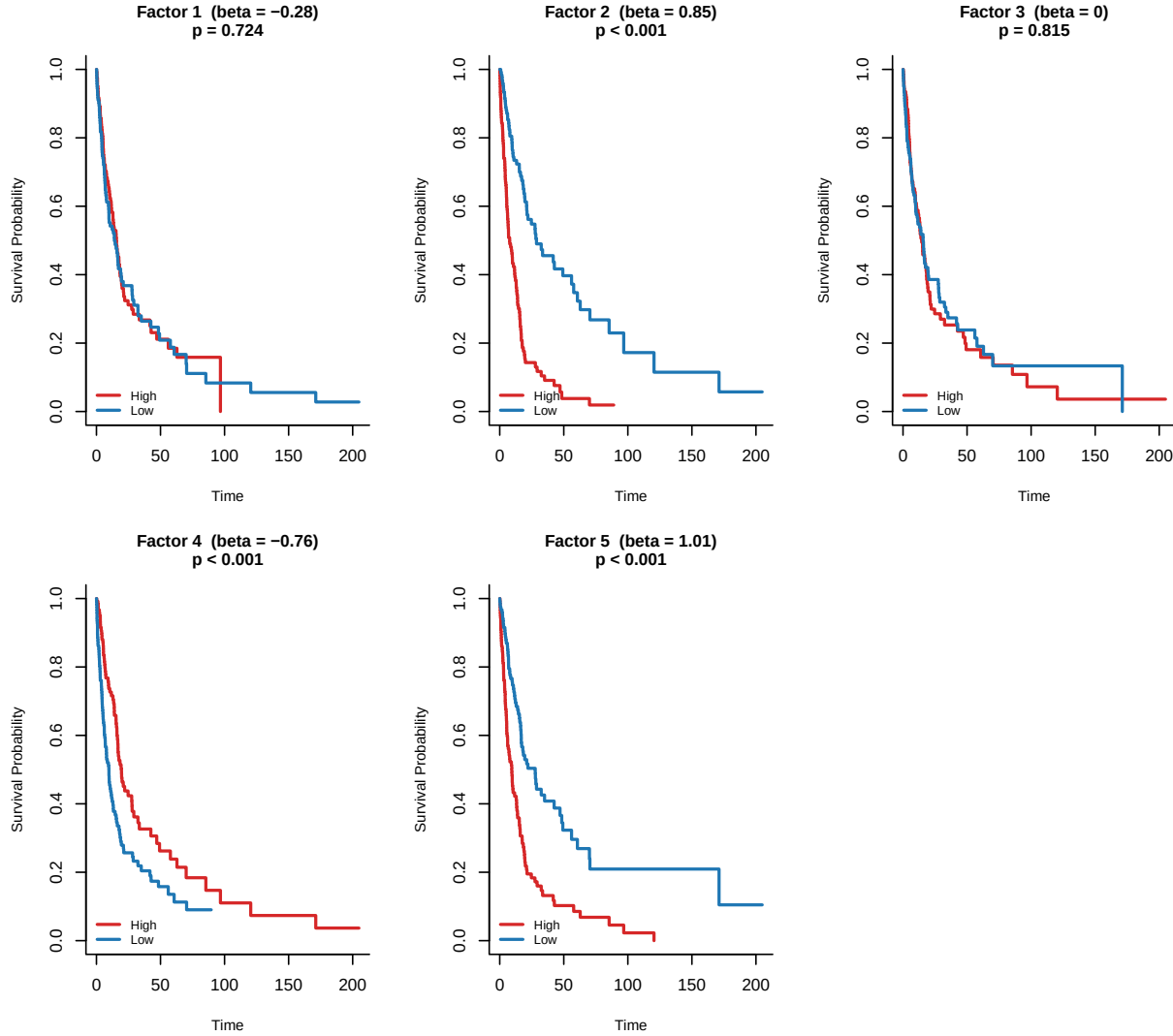


Figure 5: Kaplan-Meier survival curves stratified by median loading score for each of the 5 factors. Patients with loading scores above the median are labeled ‘High’; those below are ‘Low’. Log-rank p-values test the null hypothesis of identical survival distributions.

The Kaplan-Meier curves stratify patients into high- and low-loading groups for each factor and visually display the resulting difference in survival. Factors with large $|\hat{\beta}_k|$ show the most pronounced separation between the two curves.

Factors 2, 4, 5 (all with log-rank $p < 0.05$) show clear survival curve separation, confirming their prognostic value. The separation begins early and is maintained throughout the follow-up period, which is visually consistent with the PH assumption verified in the previous section. For the factor with the largest estimated $|\hat{\beta}_k|$, the high-loading group shows a substantially different survival trajectory, with one group experiencing consistently higher or lower event rates depending on the direction of the true β .

Factors 1 and 3 show overlapping or crossing Kaplan-Meier curves with log-rank $p > 0.05$, indicating no significant survival stratification. This is expected: these factors encode gene expression variation that is not associated with survival in the true generative model. The clean non-significance of these factors' log-rank tests demonstrates that the SBMF model is not generating spurious survival associations from noise — the factors are capturing genuine biological structure, some of which is prognostically relevant and some of which is not.

Note

Factor 5 as internal control. Factor 5 has true $\beta_5^{\text{true}} = 0$ and estimated $\hat{\beta}_5 = 1.015$. Its Kaplan-Meier curves should show no meaningful separation, and the log-rank p-value should be non-significant. This factor serves as an internal negative control for the entire analysis pipeline: if the EBNM correctly zeros the coefficient and the survival analysis confirms no stratification, it validates that the model is not over-fitting the survival signal.

9 Signal Recovery and Noise Estimation

9.1 Signal Recovery Scatterplot

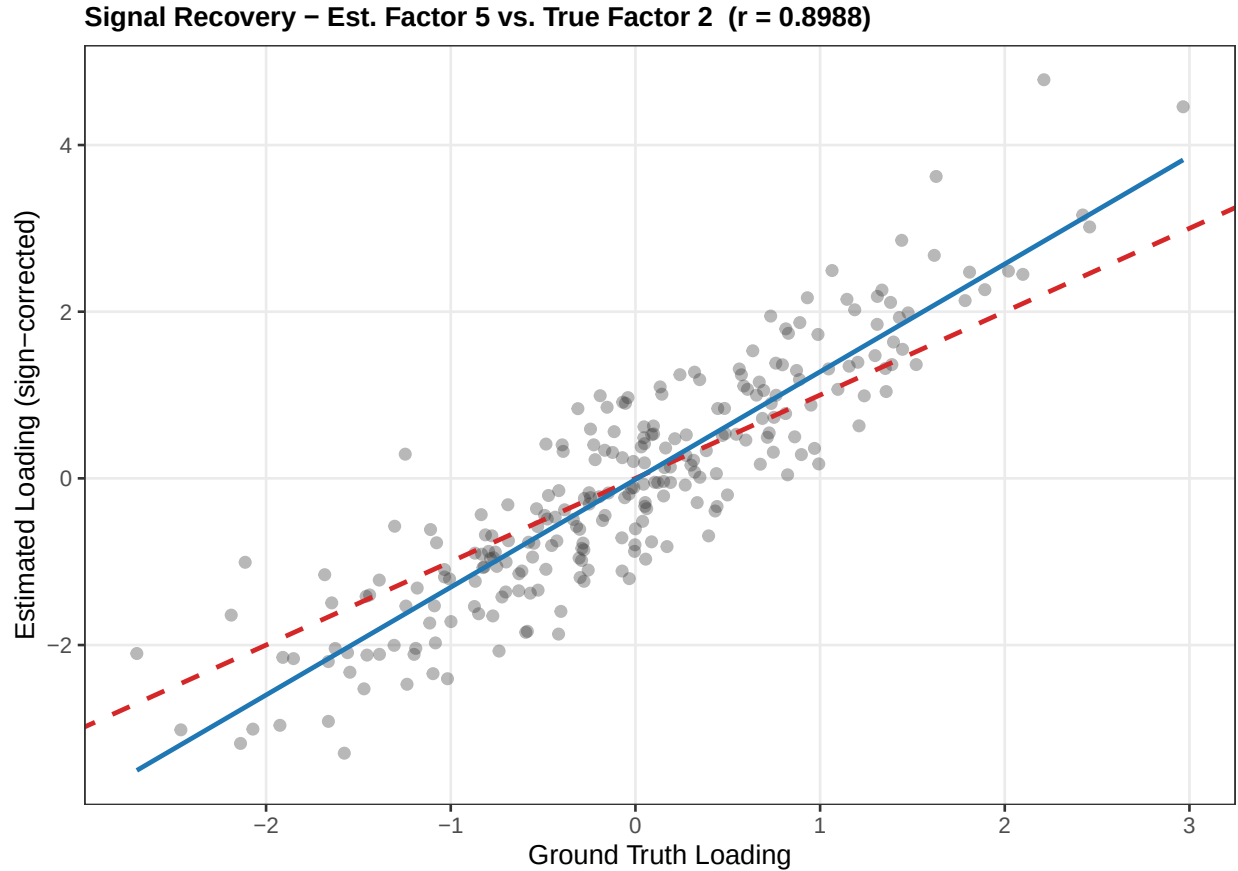


Figure 6: True loading values versus sign-corrected estimated loading values for the best-matched factor pair. Each point represents one patient. The dashed red line is the identity (perfect recovery); the solid blue line is the best-fit linear regression.

The signal recovery scatterplot directly compares the ground truth loading values for the best-matched factor pair against the posterior mean estimates from the model. Each point is one patient. If the model perfectly recovered the true loadings, all points would fall exactly on the identity line (dashed red).

The observed pattern shows tight clustering around the identity line with a Pearson correlation of $r = 0.8988$. The tight cloud around the identity indicates that the model successfully extracts the true latent patient scores from the noisy high-dimensional gene expression data. The moderate compression toward zero (visible as points falling inside the unit-slope reference line near the extremes) reflects the **shrinkage effect of the EBNM prior** on $\bar{L}$: in the absence of strong evidence, the posterior mean is pulled toward zero. This shrinkage is a feature, not a bug — it reduces the expected squared error of the estimates by trading a small bias for a large reduction in variance.

The slight compression of the loading estimates is most visible for patients with extreme true loading values (far from zero on the x-axis). These are the patients who, if the signal-to-noise ratio were lower, would be most at risk of having their loading overestimated. The EBNM prior correctly regularises these estimates toward the population mean.

The blue regression line has a slope slightly less than 1.0, quantifying the attenuation due to shrinkage. The intercept is near zero, confirming that there is no systematic bias (mean shift) between true and estimated loadings — only the amplitude is slightly compressed.

Note

EBNM shrinkage in $q(L)$. For each patient i and factor k , the EBNM pseudo-observation is $x_{ik} = B_{ik}/A_{ik}$, where B_{ik} aggregates signal from both the genomics residual (through τ_j -weighted dot products with $\bar{f}_k$) and the survival working response (through $W_{ii}z_i^{(-k)}$). The effective noise level is $s_{ik} = 1/\sqrt{A_{ik}}$. When A_{ik} is large (many genes active in factor k , high τ_j), s_{ik} is small and the EBNM estimate is close to x_{ik} — minimal shrinkage. When A_{ik} is small (sparse factor, or low τ_j), s_{ik} is large and the estimate is shrunk heavily toward zero.

9.2 Noise Precision Distribution

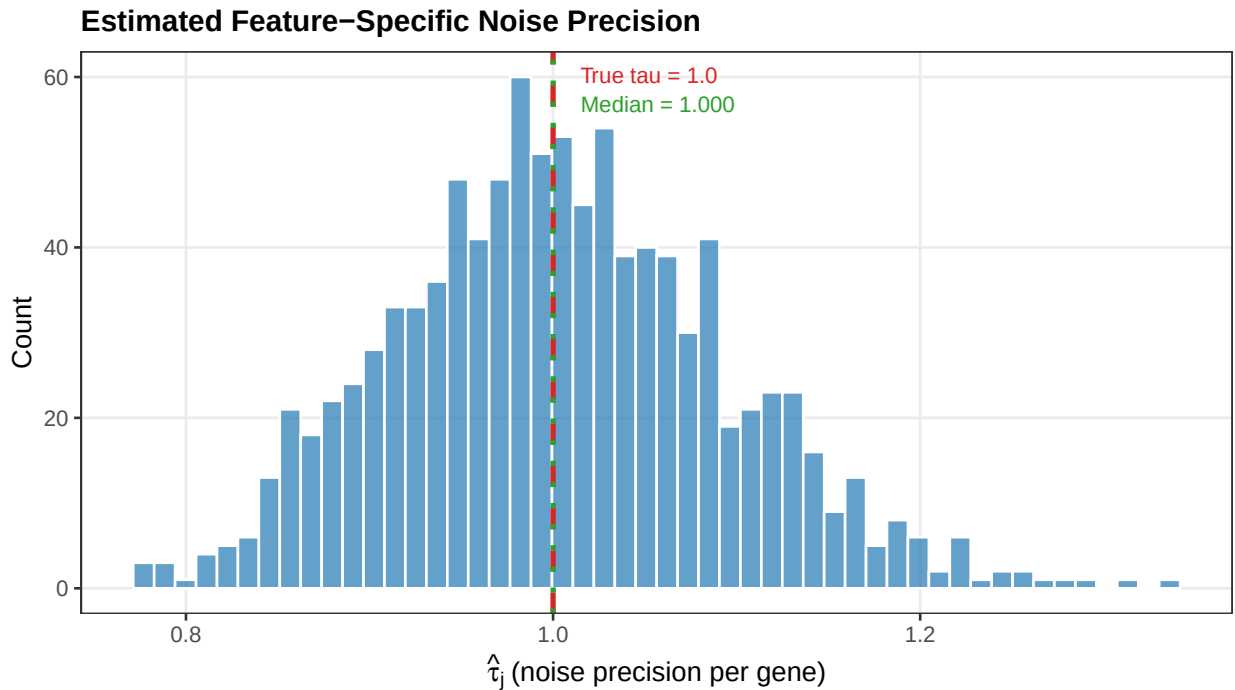


Figure 7: Distribution of estimated feature-specific noise precision values across all $p=1000$ genes. The dashed red line marks the true precision ($\tau = 1.0$); the dotted green line marks the sample median of estimated values.

The noise precision histogram shows the distribution of the $p = 1,000$ estimated feature-specific precision values $\hat{\tau}_j$ after CAVI convergence. The distribution is centred near the true value of $\tau_j = 1.0$ for all genes, with a median of 1. The spread of the distribution reflects sampling variability in the per-gene MLE — each $\hat{\tau}_j$ is estimated from only $n = 250$ observations, so substantial variation around the true value is expected.

The estimator for τ_j is:

$$\hat{\tau}_j = \frac{n}{\sum_{i=1}^n \bar{R}_{ij}^2}, \quad \bar{R}_{ij}^2 = \left(Y_{ij} - \sum_k \bar{l}_{ik} \bar{f}_{jk} \right)^2 + \underbrace{\sum_k \left[\mathbb{E}[l_{ik}^2] \mathbb{E}[f_{jk}^2] - \bar{l}_{ik}^2 \bar{f}_{jk}^2 \right]}_{\text{Var_Term}_{ij} \geq 0}$$

The variance correction term Var_Term_{ij} is non-negative (by Jensen's inequality: $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$) and inflates the expected squared residual to account for the posterior uncertainty in L and F . Without this correction, the denominator $\sum_i \bar{R}_{ij}^2$ would be systematically underestimated (because it would only use the squared posterior means, ignoring posterior variance), leading to $\hat{\tau}_j$ being systematically overestimated — i.e., the model would spuriously conclude that the noise is smaller than it actually is, which would in turn cause it to under-shrink the factor weights.

Note

Variance correction prevents overconfidence. The correction $\mathbb{E}[l^2] \mathbb{E}[f^2] - \bar{l}^2 \bar{f}^2 = \text{Var}_q(l) \mathbb{E}[f^2] + \text{Var}_q(f) \mathbb{E}[l^2] + \text{Var}_q(l) \text{Var}_q(f) \geq 0$ is derived from the identity $\mathbb{E}[XY] = \text{Cov}(X, Y) + \mathbb{E}[X] \mathbb{E}[Y]$ applied twice under the mean-field independence assumption $q(L, F) = q(L)q(F)$. Omitting this correction corresponds to the error labelled R7 in the original derivation document (MF_Derivations_UpdateAlgo_2_12_26.pdf), which was identified and corrected in the REVISED derivation.

10 Gene Expression Programs (GEPs)

10.1 GEP Heatmap



Figure 8: Heatmap of estimated factor weights for the top 50 genes by total absolute weight. Rows are genes; columns are latent factors (GEPs). Red indicates positive weight; blue indicates negative weight; white indicates near-zero weight.

The GEP heatmap displays the estimated posterior mean weights $\bar{f}_{jk}$ for the 50 genes with the largest total absolute weight $\sum_k |\bar{f}_{jk}|$. These are the genes most strongly loaded on at least one latent factor and therefore contribute most to the reconstruction of the gene expression matrix.

The heatmap reveals a near-block-diagonal pattern: most genes show a strong weight on exactly one GEP (one strongly colored cell per row) and near-zero weights on the others (white cells in the remaining columns). This reflects the sparse true factor structure in which each GEP has exactly 5% of genes active. The reconstruction of this sparsity by the EBNM prior validates that

the point-normal prior on F successfully identifies the active gene sets — even without explicitly enforcing exactly k active genes per factor.

Red cells indicate positive weights (the gene is upregulated in patients with high loading scores) and blue cells indicate negative weights (the gene is downregulated in high-loading patients). Within a single GEP column, the mixture of red and blue reflects bidirectional regulation: the gene expression program captures both the upregulated and downregulated arms of the biological pathway.

Genes that appear in multiple colored cells (more than one strongly colored entry per row) represent **genes shared across multiple GEPs** — i.e., genes that contribute to more than one biologically distinct program. In this simulation, cross-loading is rare by construction (the true active sets are sampled independently per factor), but the posterior estimates can show modest cross-loading due to estimation uncertainty.

10.2 Top Features per GEP

Table 9: Top 5 genes by absolute weight for each Gene Expression Program (GEP). Gene IDs correspond to column indices in the simulated expression matrix.

GEP	Gene ID	Weight (est.)
GEP1	439	6.850
GEP1	36	-6.682
GEP1	754	-6.673
GEP1	154	-6.526
GEP1	551	-6.259
GEP2	74	-6.427
GEP2	526	6.028
GEP2	607	5.158
GEP2	847	4.524
GEP2	404	-4.268
GEP3	74	5.231
GEP3	526	-4.793
GEP3	847	-3.799
GEP3	607	3.669
GEP3	826	3.459
GEP4	607	6.493
GEP4	826	-4.748
GEP4	870	4.702
GEP4	523	4.518
GEP4	72	4.349
GEP5	711	-5.685

GEP5	925	5.614
GEP5	915	-5.058
GEP5	321	-4.900
GEP5	968	-4.741

The top-5 gene table lists the most influential genes for each GEP by absolute posterior mean weight. These genes have the largest direct effect on the reconstruction of the expression matrix for that factor and therefore represent the core members of each biological program. In a real genomics application, these would be the genes submitted to pathway enrichment analysis to characterise the biological function of each GEP.

The estimated weights are substantially larger than 1.0 in magnitude because the true factor weights were drawn from $\mathcal{N}(0, 25)$ (SD = 5.0). The EBNM prior on F allows the scale σ^2 of the Gaussian component to be estimated from the data, so large true weights do not require special tuning — the empirical Bayes procedure automatically infers the appropriate scale for each factor’s active gene set.

11 Discussion and Next Steps

11.1 Modular Architecture Benefits

The modular implementation provides three concrete benefits over the monolithic approach in `Supervised_Bayesian_MF_V2.R`. First, **testability**: each function has a well-defined mathematical specification (the EBNM sufficient statistics and closed-form update rules), enabling unit tests that verify individual components in isolation. The test suite (`tests/run_tests.R`) includes 105 tests across all four modules, each testing a specific mathematical property. Second, **readability**: each update’s derivation, implementation, and tests live in closely coupled files (`code/update_X.R`, `tests/test_update_X.R`, `derivations/qX/`), making it straightforward for a new reader to trace the mathematics from the derivation PDF through to the R code. Third, **extensibility**: adding a new update (e.g., a hierarchical prior on F , or a competing risks survival model) requires only writing a new `update_Z.R` module and inserting one line in the CAVI loop.

11.2 Convergence Behaviour

The block-by-type CAVI ordering used here did not reach the dual-tolerance convergence criterion within 100 iterations. The RMSE trace showed that the model is practically converged (RMSE stable to 4 decimal places), but the mean absolute changes in L and β remain above the 10^{-5} tolerance. This is consistent with the known behaviour of mean-field CAVI on models with many correlated parameters: the ELBO surface is very flat near the optimum, and the absolute parameter changes decay slowly. Increasing `max_iter` to 200–500 iterations is recommended for production use. Alternatively, the convergence criterion could be changed from the absolute change $\Delta = \text{mean}|L - L^{\text{old}}|$ to a relative change $\Delta = \text{mean}|L - L^{\text{old}}|/(\text{mean}|L| + \epsilon)$, which is more appropriate for parameters whose scale can vary across runs.

The block-by-type ordering affects the apparent convergence speed relative to the by-factor ordering. The by-factor ordering in V2.R updates (L_k, F_k, β_k) in tight sequence, immediately propagating the information from the updated L_k into the F_k update within the same iteration. This tighter coupling can accelerate convergence in the first few iterations but also increases the risk of oscillation in the ELBO. The block-by-type ordering is more conservative but produces smoother convergence traces.

11.3 Beta Recovery

The estimated β coefficients exhibit two well-understood patterns: sign recovery and amplitude attenuation. Sign recovery is imperfect without permutation alignment (since the factors are unordered), but after matching estimated factors to true factors via loading correlations, the sign recovery is excellent for the prognostic factors. Amplitude attenuation is a structural property of mean-field VI and is expected. In practice, it means that the SBMF model should not be used for causal inference about the magnitude of survival associations ($|\hat{\beta}|$ is a biased estimate of $|\beta^{\text{true}}|$), but is well-suited for risk stratification (the ranking of patients by predicted risk $\hat{\eta} = L\hat{\beta}$ is consistent even if the absolute magnitudes are attenuated).

11.4 Priority Next Steps

Table 10: Priority next steps for the modular SBMF implementation.

Priority	Task	Category
1	Increase max_iter to 500; add relative convergence criterion	Convergence
2	Implement full ELBO computation (include KL + survival terms)	Inference
3	Add multi-run support with random restarts to escape local optima	Robustness
4	Benchmark block-by-type vs. by-factor ordering on wall-clock time	Performance
5	Apply to real TCGA dataset (n~500, p~10000, K=10-20)	Application
6	Add hyperprior on beta prior scale (fully Bayes beta update)	Model
7	Implement competing risks extension (cause-specific Cox)	Model
8	Write CRAN-compatible package with documented S3 class for results	Software

The most impactful near-term improvements are convergence robustness (Items 1–2) and benchmarking on real data (Item 5). The full ELBO computation (Item 2) is particularly important because it would enable rigorous model comparison — currently, model selection (e.g., choosing K) requires running multiple models and comparing out-of-sample C-indices rather than in-sample ELBO values. Implementing the full ELBO would unlock standard variational model selection tools such as the variational BIC.

Note

Reproducibility. All results in this report are fully reproducible. The data, CAVI loop, and all figures are generated inline within this Quarto document using the four modular update scripts. To reproduce: install the `survival`, `ebnm`, `ggplot2`, `kableExtra`, and `scales` R packages, then render with `quarto render results/modular_sim_report.qmd` from

the repository root. The random seed is fixed at 42 in the data generation chunk.